

# 1. SIGNAL GENERATION AND BASIC SIGNAL OPERATIONS

## Objective

This experiment is divided into four simulations:

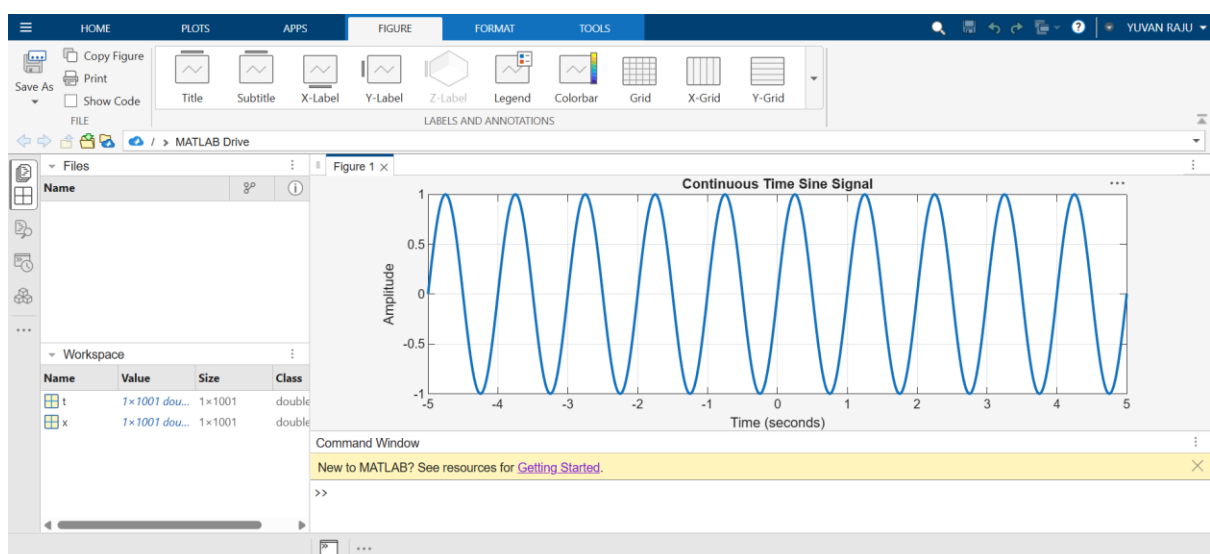
1. Generation of Continuous Time Signals
2. Generation of Discrete Time Signals

### Experiment 1.1 Generation of Continuous Time Signals

#### CODE:

```
clc;  
clear;  
close all;  
  
% Time axis t = -5:0.01:5;  
  
% Continuous-time sine signal x = sin(2*pi*t);  
  
% Plotting figure;  
  
plot(t, x, 'LineWidth', 2);  
  
grid on;  
  
title('Continuous Time Sine Signal');  
  
xlabel('Time (seconds)');  
  
ylabel('Amplitude');
```

#### OUTPUT:



## Experiment 1.2 Generation of Discrete Time Signals

### CODE:

```
clc;

clear;

close all;

% Discrete-time index n = -10:10;

% Discrete-time sinusoidal signal x = sin(0.2*pi*n);

% Plotting figure;

stem(n, x, 'filled');

grid on;

title('Discrete Time Sinusoidal Signal');

xlabel('Sample Index (n)');

ylabel('Amplitude');
```

### OUTPUT:

